

Neuro-Symbolic Verification of LLM Outputs for Data-Sensitive Domains

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Abstract. LLMs deployed in high-stakes domains face fundamental reliability challenges: hallucinations and privacy vulnerabilities create unacceptable risks when errors carry legal, financial, or safety consequences. This paper presents a validated systems architecture that combines tableaux-based input verification, actor-based fault isolation, and type-aware post-generation verification, using deterministic symbolic checks for structured claims and neural semantic similarity for free-text claims. Validated through HAIMEDA, a locally deployed medical-device damage assessment system, the architecture detects 83% of hallucinated structured entities and 72% of semantic fabrications while reducing report creation time by 30%, demonstrating that neuro-symbolic architectures can provide principled safeguards for LLM deployment in data-sensitive domains.

Keywords: Neuro-symbolic AI · LLM verification · Formal verification · Trustworthy AI · Hybrid AI · Actor model · AI system architecture

1 Introduction

Large language models (LLMs) perform strongly across many tasks, yet their deployment in high-stakes domains remains challenging. Hallucinations, i.e., confidently generated but factually unsupported content, occur at rates of 3% to 10% even in state-of-the-art models [12,13]. In professional settings such as medical diagnosis, legal review, and regulatory affairs, where errors carry legal, financial, or safety consequences, such failure rates are unacceptable.

Beyond accuracy, many data-sensitive workflows cannot rely on cloud-based APIs due to regulatory, institutional, or externally imposed restrictions on data transmission [7,14]. Even when GDPR-compliant data processing agreements are available, confidentiality duties and contractual obligations arising from medical-legal expert practices may still preclude external transmission. Although retrieval-augmented generation (RAG) and chain-of-thought prompting improve reliability, they do not provide the guarantees required for high-stakes workflows. A more principled approach requires integrating complementary AI paradigms.

Hybrid AI systems address this gap by combining neural methods, which excel at pattern recognition and natural language processing, with symbolic

methods that provide explicit reasoning and verifiable guarantees [9]. Despite the growing recognition of this potential, there is still a lack of validated end-to-end verification architectures for locally deployed LLM systems in data-sensitive domains [3].

This paper makes two technical contributions for verifiable LLM deployment in data-sensitive domains. First, we propose a verification architecture for LLM-assisted generation in data-sensitive domains, combining tableaux-based input validation, actor-based fault isolation, and a type-aware post-generation verification method that uses deterministic symbolic checks for structured claims and semantic similarity scoring for free-text claims. Second, we validate this architecture through HAIMEDA, a locally deployed medical device assessment reporting system that demonstrates strong detection of unsupported content and measurable improvements to workflow. The full architectural details, complete validation procedure, and extended evaluation are available in the extended online version [22].

2 Related Work

Research relevant to hybrid verification of LLM-generated content builds on three areas: neuro-symbolic architectures, hallucination detection methods, and output verification systems.

Neuro-symbolic integration combines neural learning with symbolic reasoning [9]. Representative frameworks such as Logic Tensor Networks [2] and Deep-ProbLog [16] embed logic into differentiable or probabilistic pipelines, but they require architecture changes or joint training, making them less suitable as external verification layers for pretrained LLMs.

For hallucination detection, self-consistency methods [24] measure agreement across sampled responses, while retrieval-augmented verification [8] cross-references outputs against external knowledge. Both approaches can inherit distributional biases from the models they verify [12,13]. Runtime guardrails such as Guardrails AI [5] and NeMo Guardrails [19] enforce format and policy checks. Similarly, domain-specific systems such as TrustKG [4] demonstrate verification in specialized contexts. However, these approaches are typically point solutions rather than generalizable architectural patterns. Notably, none combines formal input verification with type-aware semantic post-processing into a complete architecture that can be deployed under strict data-sovereignty constraints.

3 Verification Architecture and Design Rationale

Deploying LLMs in data-sensitive domains raises verification challenges that guided the design of the proposed architecture. Input validation benefits from decidable guarantees before committing to costly inference. Output verification requires mechanisms for both symbolic and neural analysis, ideally with fault isolation to prevent cascading failures. Privacy considerations in regulated domains often necessitate local deployment rather than cloud-based APIs. Our architecture

addresses these challenges by combining tableaux-based verification for input constraints, actor-model concurrency for fault-isolated parallel processing, and neuro-symbolic integration for output verification.

The architecture is guided by four design principles: local-first processing for data sovereignty [14], modular separation of symbolic and neural components [18], actor-based fault isolation with functional immutability for robustness [1,11], and type-aware claim verification, i.e., deterministic symbolic checks for structured claims and neural semantic checks for free-text claims [9].

3.1 Tableaux Logic for Input Verification

In workflows for report generation in data-sensitive domains, input validation is a consistency problem over interacting constraints. Report sections have different requirements: some are narrative, some follow a strict schema, and some carry legally consequential obligations. This variability requires a mechanism that can express cumulative obligations and alternative sufficiency paths within one declarative scheme.

Tableaux methods decompose formulas into conjunctive (α) obligations that must all hold, and disjunctive (β) branches that represent admissible alternatives. For propositional and decidable fragments, this yields terminating, sound and complete procedures [23]. The architecture encodes validation as a configurable hierarchy: *core conditions* $\mathcal{C}$ evaluate atomic predicates such as metadata presence or title quality; *meta conditions* express set-theoretic relationships over condition groups; and *aggregate conditions* compute statistics over condition attributes. Let $\mathcal{C}_{pos}$ collect conditions whose observed satisfaction state matches their expected state, and let $\mathcal{C}_{neg}$ collect the remaining polarity mismatches. A mandatory-consistency gate checks $(\mathcal{C}_{req} \cap \mathcal{C}_{eval}) \subseteq \mathcal{C}_{pos}$, i.e., all evaluated required conditions must match their expected state, while an aggregate rule bounds mismatch via $|\mathcal{C}_{neg}| \leq \tau$. Together, these condition types, dependency checks, and gating rules are combined into a single validation algorithm described in full in the extended online version [22].

Example: Context-Sufficiency Gate. As a concrete example from medical-device assessment, before generating the section that describes the damage, the system checks whether the input provides sufficient grounding to avoid hallucinated device specifics. A meta condition c_{ready} accepts two structurally distinct evidence combinations:

$$\varphi_{c_{ready}} \equiv (c_{dev_type} \wedge c_{serial} \wedge c_{category}) \vee (c_{category} \wedge c_{damage_desc} \wedge c_{party}),$$

where the left disjunct captures full technical identification and the right captures legally sufficient narrative grounding. The tableau opens one branch per disjunct. If the serial number is absent, the first branch fails, but the second can still satisfy c_{ready} if the narrative evidence is sufficient. Generation is blocked only if both branches fail. This avoids enumerating all admissible field combinations procedurally and illustrates how α/β -decomposition maps directly to set intersection and union over condition identifiers.

3.2 Type-Aware Post-Processing Verification

LLMs may produce text that satisfies formatting constraints while lacking source support [13]. Neural self-verification is unreliable because it can inherit the same distributional biases that produce hallucinations [12]. The architecture therefore applies a type-dependent strategy. Structured entities such as dates, identifiers, and numeric values undergo symbolic pattern matching for deterministic verification. Semantic content is assessed with a five-metric similarity scheme combining TF-IDF similarity, domain-specific semantic similarity, normalized Euclidean similarity, token overlap, and keyword overlap. Multilingual sentence embeddings [20] support the semantic component. Inter-metric agreement yields a confidence score. Bidirectional comparison detects two failure modes: *hallucinations* (output entities unsupported by input) and *missing content* (input entities absent from output). Detected discrepancies trigger interactive corrections.

For example, if the source contains BF-1HTJ0 and 2024-11-03, while the generated output contains BF-1HTJ0 and 03.11.2024, symbolic matching accepts the date as format-equivalent but flags the identifier mismatch deterministically. Statement-level claims are evaluated using the five-metric scheme with unresolved cases forwarded for manual oversight.

4 HAIMEDA: System Instantiation

Medical device damage assessment combines PII, health-related data, and legally consequential documentation, making it a representative data-sensitive workflow. HAIMEDA (Hybrid AI for Medical Device Assessment) instantiates the proposed architecture in this setting. While cloud APIs can be made GDPR-compliant via data-processing agreements [7], this case is constrained more strictly. HAIMEDA was developed in collaboration with a domain expert who is a publicly appointed and sworn expert witness and a certified medical device engineer. This expert works under statutory and contractual confidentiality obligations. Requirements arising from the Medical Devices Regulation [6], its German implementing act [10], and contracts with commissioning authorities therefore make local-first processing a deployment requirement rather than a preference.

4.1 System Architecture

HAIMEDA is coordinated by a central orchestration layer that runs generation and verification in fault-isolated workers. The verification workflow centers on the interaction between two core modules. The *Report Authoring Module* (RAM) handles report generation through structured templates and incorporates verification feedback during revision. The *Information Integrity Verification Module* (IIVM) implements multi-stage verification through the symbolic pre-processor (Section 3.1) and the type-aware post-processor (Section 3.2). Additional modules support RAG-based information retrieval, feedback presentation, and dynamic interactive editing. Figure 1 illustrates module coordination during report creation.

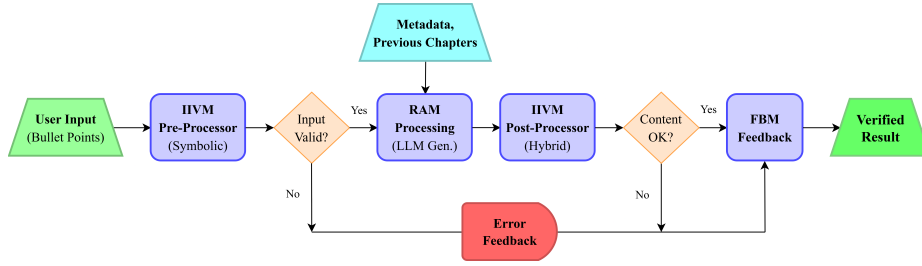


Fig. 1. HAIMEDA workflow. The symbolic pre-processor validates input, the orchestration layer gates or permits generation, the post-processor verifies output, and the Feedback Module (FBM) returns validation results to the user.

5 Evaluation

All experiments ran on consumer hardware representative of deployment conditions: AMD Ryzen 9 5900X with 64 GB of RAM, and an NVIDIA RTX 4090 with 24 GB of VRAM. Content generation used a fine-tuned **Llama3-DiscoLeo-Instruct-8B** variant that was trained on 899 historical assessment reports, all on local hardware. The IIVM was evaluated over 100 runs per component using three test suite categories: correct statements serving as negative controls, deliberately hallucinated content with substituted entity values and plausible but incorrect statements, and versions with critical information removed. Ground truth was established through manual annotation against the source documents.

Practical utility was assessed together with the domain expert through a controlled task-timing comparison [21]: three standardized assessments were completed using HAIMEDA and three without it. Standardization was achieved by selecting comparable cases, which were evaluated using uniform assessment templates defined by the domain expert. The scope of the measurement included content writing, document-based research, and final review. Unaffected activities, such as on-site inspection and stakeholder communication, were excluded. The 30% reduction in total productive time reflects the mean across the six timed assessments. Content writing time decreased by 50%, document-based research by 33%, and final review time doubled. These findings align with previous studies demonstrating that generative AI can significantly enhance the productivity of writing-intensive tasks, while necessitating human verification and integration efforts to ensure professional-grade output [17,15].

5.1 Verification Effectiveness

Table 1 reports the component-wise detection rates for unsupported and missing content across the verification pipeline, highlighting the complementarity of symbolic and semantic methods. Here, “consensus similarity” refers to the multi-metric semantic verification scheme described in Section 3.2.

Table 1. IIVM verification performance across component types, showing detection rates for hallucinated and missing content based on 100 test runs per component.

Component	Method	Hallucinated	Missing
Preprocessing	Tableaux logic	—	100%
Entities	Regex patterns	83%	90%
Phrases	Semantic rules	76%	90%
Statements	Consensus similarity	72%	65%

Overall, the results support the type-aware design, whereby symbolic checks provide deterministic coverage when constraints are formalizable, while neural similarity extends verification to unconstrained text [12,13]. However, this broader semantic coverage comes at a latency cost: while symbolic preprocessing stays below 20 ms, statement verification averages 16.6 s for chapters of about 600 words. This discrepancy reflects the greater difficulty of verifying semantically unconstrained text. Structured entities are discrete and can often be matched exactly. In contrast, hallucinated statements may be paraphrased, appear plausible in context, or be only partially supported. Detecting omissions is even more difficult because the verifier must infer the absence of expected content from weak similarity rather than from an explicit contradiction.

The observed performance depends on domain-specific resources, regex patterns, and threshold calibration for statement-level scoring. Since no dedicated ablation study was conducted, the evaluation targets integrated system utility rather than the effects of isolated components. Nevertheless, the component-wise results support the usefulness of the type-aware split, as rule-only checks cannot reliably cover free-text statements, whereas embedding-only verification lacks the exactness guarantees required for structured fields such as identifiers and dates.

6 Conclusion

This paper presented a neuro-symbolic architecture with a type-aware verification approach for LLM-assisted generation in data-sensitive domains. The principled decomposition assigns symbolic methods to constraint checking and hybrid neuro-symbolic methods to semantically unconstrained content, enabling fault isolation and locally deployable quality control pipelines. The HAIMEDA instantiation demonstrates the practicality of this decomposition in a regulated professional workflow. It detects 83% of hallucinated structured entities and 72% of semantic hallucinations while reducing report creation time by 30%. Beyond detection rates, the architecture transforms verification signals into actionable feedback that facilitates interactive report revision.

Future work should evaluate its transferability to legal and financial settings, conduct ablation studies over alternative constraint engines and similarity backends, and assess whether larger, locally deployable models reduce the verification burden or if the need for guardrail architectures remains unchanged.

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